Achyuth Kolluru

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EDUCATION

University of California Berkeley, CA **January 2024 - May 2025**
M.S. Majors in Information and Data Science *Berkeley, CA*

University of California Merced, CA **August 2019 - May 2023**
B.S. Majors in Computer Science and Engineering *Merced, CA*

- Top Finisher for Innovate to Grow Event for 2022 Fall Software Engineering Capstone
- 3rd place winner for the Water Hack Challenge 2023 Issued by Secure Water Future

WORK EXPERIENCE

TheQwikGroup (TQG) & Affiliated Stealth-Mode AI Startup **September 2025 - Present**
Co-Founding Machine Learning Engineer *Palo Alto, CA*

- Developed **reinforcement learning systems for humanoid robotics** using PyTorch and Isaac Lab, developing policies from simulation through training, evaluation, and real-time deployment
- Trained high-frequency RL policies using **PPO and distributed multi-GPU training on GCP**, scaling experiments across **16 GPUs and 3,000+ parallel simulation environments**.
- Designed **observation/action spaces and reward functions** incorporating joint states, velocities, orientations, contact information, and camera/LiDAR perception, while tuning termination and reset strategies for locomotion, grasping, lifting, balance, and coordinated upper-body control.
- Ran **100K+ environment interaction steps**, analyzing reward curves, episode performance, and failure cases to identify policy instability, oscillation, falls, and unsuccessful object interactions.
- Optimized large-scale RL training through **parallel rollouts, memory-efficient batching, GPU profiling, and simulation bottleneck analysis**, improving utilization across distributed training workloads.
- Built perception-to-control pipelines combining **camera-based object detection and LiDAR localization** with learned policies, enabling real-time object-aware robotic control.

DeepLM (stealth-mode AI Startup) **January 2025 - August 2025**
Founding Machine Learning Engineer *San Jose, CA*

- Built machine learning infrastructure and high-throughput data pipelines from the ground up, supporting large-scale model training and low-latency inference workloads.
- Designed efficient data storage and retrieval workflows using **LMDB**, optimizing data access for large-scale machine learning workloads.
- Developed **Chorus**, an internal inference orchestration framework that dynamically matched ML workloads to available GPU compute and memory resources.
- Implemented production monitoring and alerting using **Prometheus and Grafana**, tracking inference health, training performance, latency, resource utilization, and system bottlenecks.
- Managed Linux-based ML infrastructure using Ubuntu, SSH, TMUX, Git, and Docker while debugging CUDA, GPU, memory, and system-level issues.
- Analyzed model and infrastructure performance metrics to identify bottlenecks and improve reliability of training and inference workflows.

University of California Merced **November 2022 – May 2023**
Research Assistant *Merced, CA*

- Developed a deep learning-based video frame-rate upsampling system using cubic interpolation and jerk derivatives, integrating **PWC-Net optical flow** to improve reconstruction quality by 20% PSNR.
- Built scalable **PyTorch** pipelines for large-scale image synthesis using multi-GPU Azure clusters and Kubernetes, processing **100K+ video frames** efficiently.
- Applied deep learning, numerical methods, and quantitative evaluation to compare model performance against existing approaches.

LANGUAGES & TECHNOLOGIES & INTERESTS

- **Languages:** Python; R; MATLAB; SQL; C++;
- **Technologies:** IsaacSim; TensorFlow; PyTorch; Scikit-Learn; Keras; Git; ROS; Linux; Jupyter; Pandas; Matplotlib; PostgreSQL; Redis; TMUX; Docker; GCP; Hadoop; PySpark; Sciy
- **Interests:** Swimming; Longboarding; Video Games; Hiking; Formula 1; DND

RELEVANT PROJECTS

Capstone Project: "Brooke" AI Real Estate Platform (link) **January 2025 – April 2025**

- Built an end-to-end AI powered real estate assistant using Python, Gradio, LangChain (Chroma + HuggingFaceEmbeddings), and RetrievalAugmented Generation to deliver a specialized chatbot, interactive budget calculator, and dynamic market trends dashboard.
- Implemented scalable data pipelines with Pandas, Altair, and Folium (plus Geopy) to generate realtime graphs and comparative maps based on user selected metros, cities, and neighborhoods—fully pre-populated from a custom questionnaire.
- Outperformed open source benchmarks in both usefulness and safety on 2,000+ real estate inquiries, demonstrating cost effective, accurate guidance that rivals traditional buying agents.

Hybrid Sentiment-Driven Stock Forecasting **January 2025 – April 2025**

- Finetuned BERT and FinBERT on 68K+ labeled tweets with Optunaguided hyperparameter search (epochs, LR, dropout, warmup, clipping), achieving 71% validation accuracy in sentiment classification.
- Extracted CLS/mean/maxpooled embeddings from tweets, aggregated daily by ticker, and merged with 10year Yahoo Finance returns and technical indicators to create a unified forecasting dataset.
- Built and compared Dense and LSTM neural networks on priceonly, sentimentonly, and combined feature sets —realizing a 56.8% directional accuracy with LSTM+meanpooled embeddings and a 15% MAE reduction over priceonly baselines.

Flight Delay Prediction Analysis **August 2024 – December 2024**

- Designed a machine learning-based system for predicting flight delays using a 30GB historical flight dataset, implementing preprocessing steps such as handling missing values, encoding categorical variables, and feature engineering to optimize model performance.
- Applied Random Forest, XGBoost, and Linear Regression, evaluating models using metrics like precision, recall, and F1 score to ensure high predictive accuracy.
- Leveraged PySpark ML on Databricks with Azure-hosted CPU-based clusters for scalable, parallelized processing and model training, ensuring efficient handling of large datasets for real-time predictions.

Image Classification with Transfer Learning **May 2024 – June 2024**

- Applied transfer learning using VGG16 and ResNet50 to classify images from the CIFAR-10 dataset, achieving 85% classification accuracy, exceeding standard benchmarks for the dataset.
- Enhanced model efficiency by fine-tuning pre-trained weights, reducing training time by 30%
- Developed a pipeline to handle batch predictions for large datasets, enabling the solution to scale to real-world applications like large-scale content moderation.
- Conducted hyperparameter optimization on SageMaker to maximize training efficiency and validate results with multi-class confusion matrix metrics.

Time Series Forecasting of CO_2 Emissions Trends **June 2024 – July 2024**

- Applied ARIMA and SARIMA models for CO_2 emissions trend analysis. Improved forecast reliability through data preprocessing, outlier detection, and anomaly correction.
- Quantified seasonality effects and refined forecasting strategies, providing actionable insights aligned with global carbon emissions reduction goals.